

JYOTHIR RAGHAVALU BHOGI

Generative AI Engineer | LLMs, RAG & AI Agents | Python, LangChain, LangGraph, FastAPI, AWS
jyothirraghavalu369@gmail.com | +91 6281073800 | [LinkedIn](#) | [GitHub](#)

Generative AI Engineer building production-oriented LLM, RAG, and agentic AI systems — RAG pipelines with embeddings and vector search, multi-step agent orchestration, LLM evaluation, and the backend APIs that expose all of it. B.Tech CSE candidate (2026) with two AI/ML internships and three shipped GenAI projects behind REST APIs.

EDUCATION

Lovely Professional University

Bachelor of Technology, Computer Science & Engineering

Jalandhar, India
July 2022 - May 2026

SKILLS SUMMARY

Generative AI: LLMs, RAG, Prompt Engineering, Embeddings, Vector Search, Semantic Search

AI Agents / Orchestration: LangChain, LangGraph, Agentic Workflows, Task Decomposition, Checkpointing

AI/ML: Machine Learning, Deep Learning, ML Pipelines, Model Inference, Model Evaluation, PyTorch, Hugging Face

APIs & Deployment: FastAPI, REST APIs, Model Serving, AWS, Docker, CI/CD, Model Deployment

Data: Python, SQL, PySpark, Apache Spark, Data Preprocessing, Feature Engineering, ETL, Batch Processing

Engineering: Performance Optimization, Debugging, Workflow Automation, Git, GitHub

WORK EXPERIENCE

Generative AI Engineer Intern | Know AI

June 2026 - August 2026

- Built Generative AI workflows using LLM inference, NLP, prompt engineering, and structured model evaluation.
- Designed RAG pipelines using embeddings and vector search to improve contextual retrieval and AI reliability.
- Built backend services and REST APIs (FastAPI) integrating LLMs, AI inference, search, and automation workflows.
- Developed AI/ML evaluation pipelines for model testing, error analysis, output validation, and performance analysis.
- Conducted model evaluation, error analysis, and output validation that directly improved ML reliability.

Data Engineer Intern | flickDONE

June 2025 - August 2025

- Built Python-based ML data pipelines processing 50,000+ records for preprocessing, training, and validation.
- Engineered automated preprocessing and data validation workflows, reducing manual verification by 70%.
- Optimized AWS ML pipelines, improving execution speed by 25% through bottleneck analysis.
- Developed reusable AI workflows for conversational AI, document processing, and business automation.
- Collaborated with engineering teams to improve scalability, reliability, and maintainability of ML platforms.

PROJECTS

Production-Grade RAG Engine

June 2024 - August 2024

Python, FastAPI, LangChain, Qdrant, AWS

- Built an end-to-end production RAG pipeline: document ingestion, chunking, embedding, vector retrieval, and context generation.
- Implemented semantic search with a Qdrant vector database for efficient document context retrieval.
- Added reranking and retrieval evaluation to improve context relevance and response quality.
- Improved system reliability through fallback execution, testing, monitoring, and error handling.

LangGraph Workflow Orchestrator

August 2025

Python, LangGraph, LangChain, FastAPI, Gemini

- Built a stateful AI agent orchestration platform with multi-step workflows, state management, and conditional routing.
- Developed modular components for task decomposition, tool execution, and automated agent coordination.
- Exposed workflow execution through FastAPI REST APIs, enabling scalable, reusable AI task automation.
- Improved workflow reliability through checkpointing, failure handling, testing, and execution monitoring.

Regulatory Compliance RAG Assistant

April 2026 - June 2026

Python, Streamlit, LangChain, Embeddings, Vector Search

- Built a Generative AI compliance assistant with document ingestion and semantic retrieval pipelines.
- Implemented context-grounded generation to deliver relevant and reliable compliance responses.
- Developed interactive document upload and querying workflows using Streamlit.
- Improved response reliability through retrieval testing, evaluation, debugging, and quality checks.

CERTIFICATES

Generative AI with Large Language Models — Coursera

March 2024

- Applied prompt engineering techniques for production LLM applications.
- Built retrieval workflows, AI output evaluation, and RAG pipeline performance validation.

Python, Data Science & Machine Learning — CipherSchools

August 2024

- Built machine learning models using PyTorch and applied ML techniques for model development and evaluation.
- Developed Dockerized environments supporting reproducible experimentation and deployment.